

Recursive self-improvement

A mechanism-level view

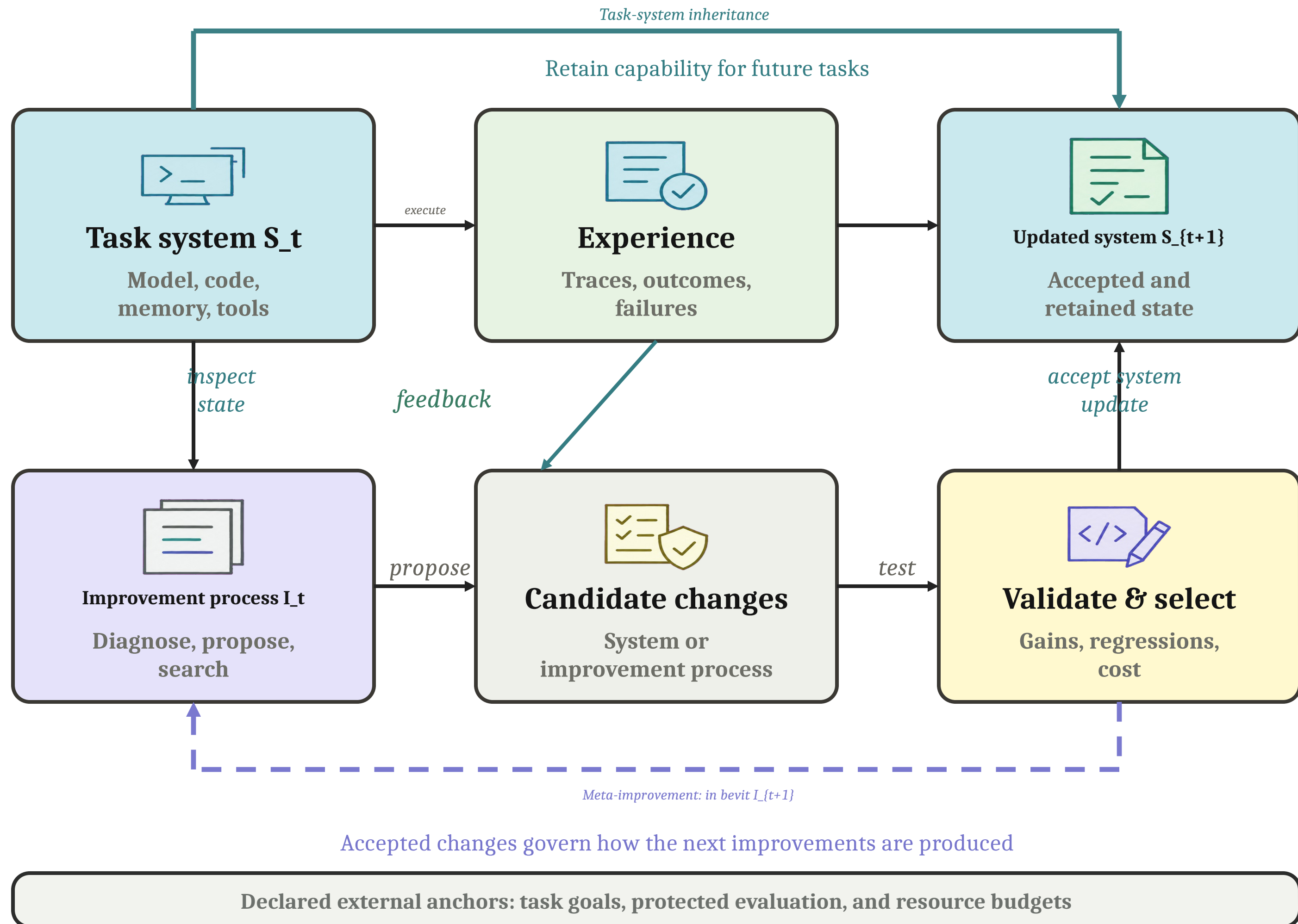
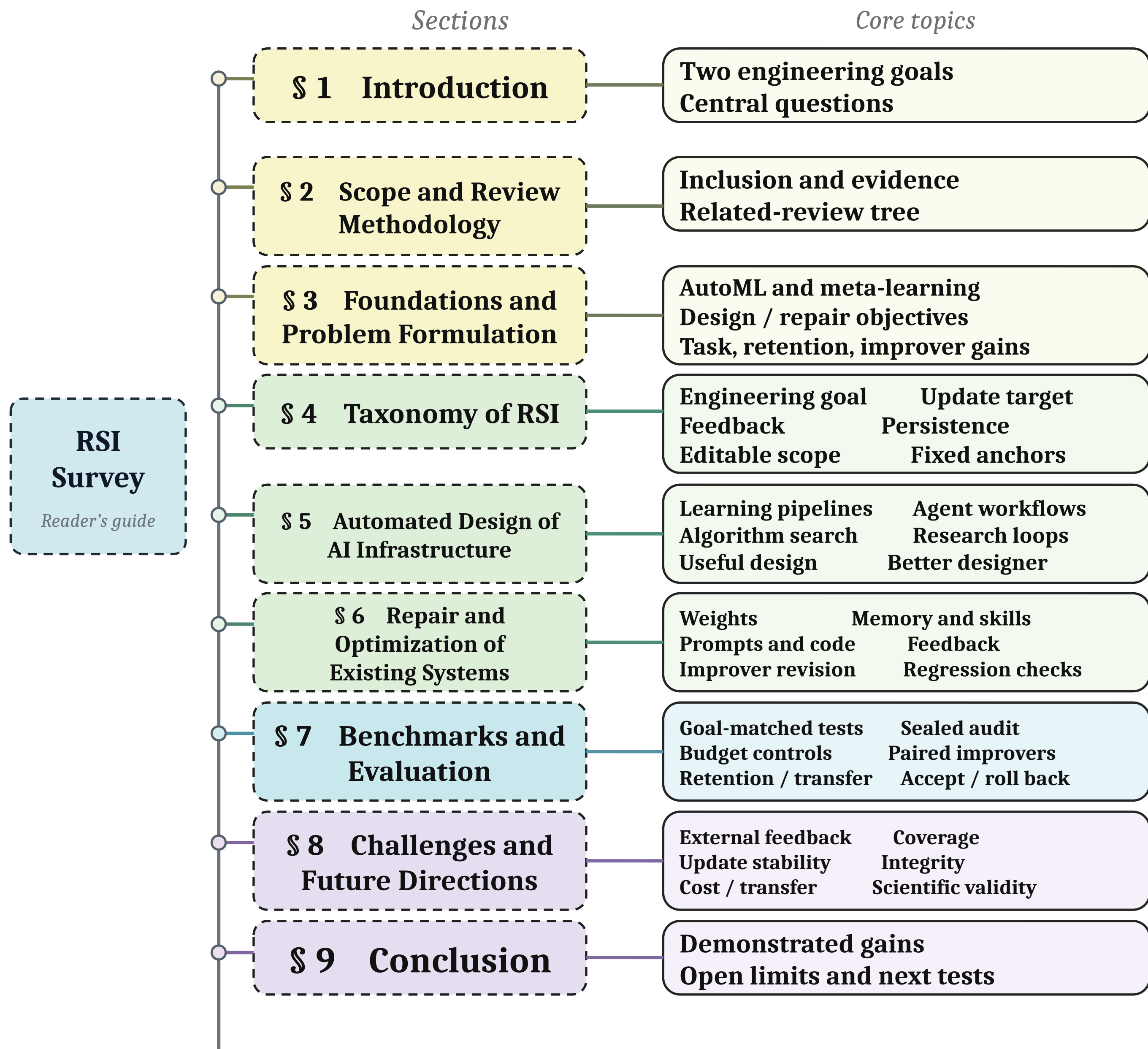


Figure 1. A mechanism-level view of recursive self-improvement. Experience informs candidate changes. Validation determines what is retained: the upper loop carries task-system state into later tasks; the dashed lower loop carries changes to the process that produce later improvements. External anchors specify the conditions outside the update scope. The diagram describes possible dependencies, rather than a guarantee of gains.



The landscape of recursive self-improvement

Foundations, enabling mechanisms and emerging systems

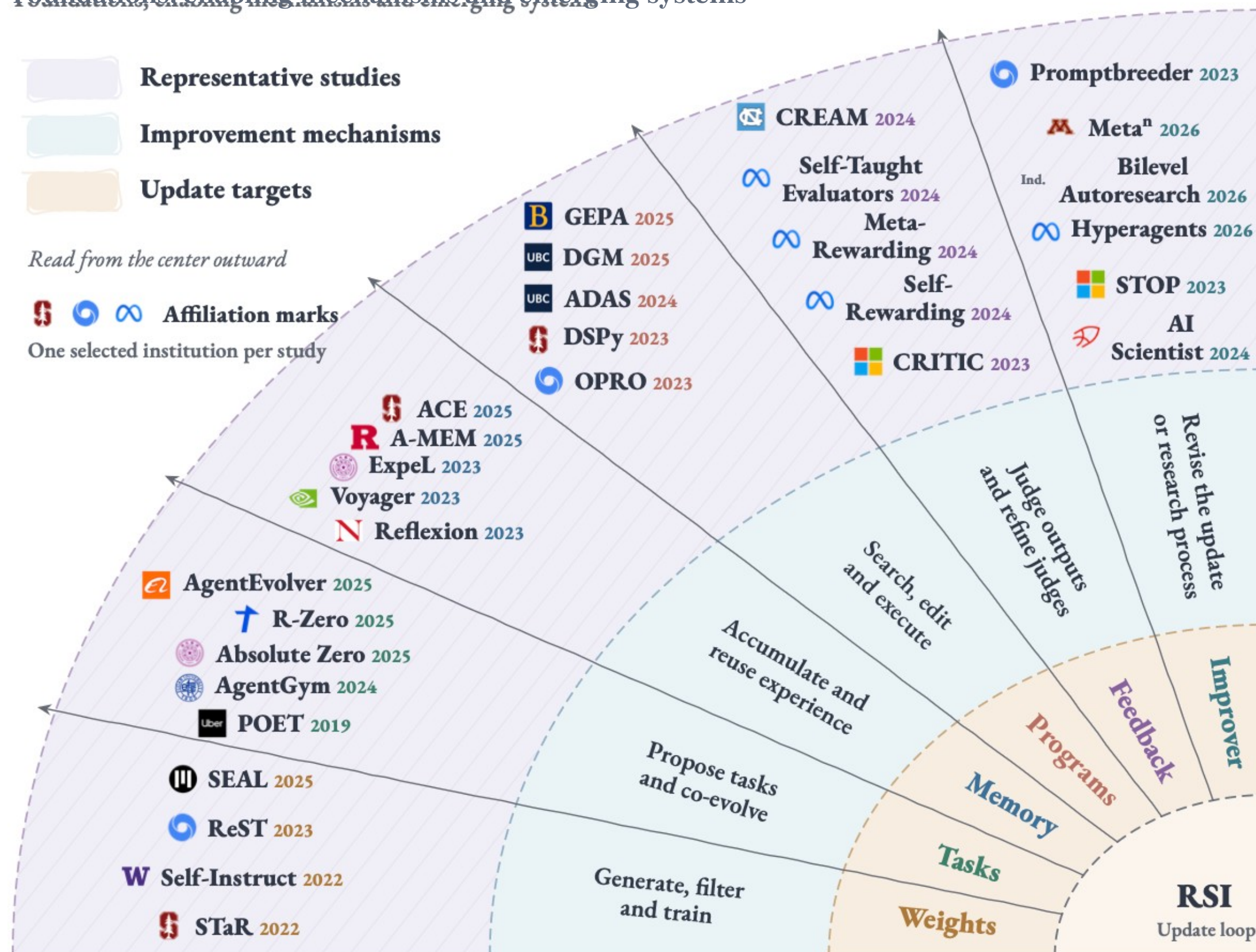


Figure 3. Landscape of mechanisms relevant to recursive self-improvement. The map connects update targets in the inner band, improvement mechanisms in the middle band, and representative studies in the outer band. The six sectors cover weights, tasks, memory, programs, feedback, and the improvement process. A work may involve several targets; its placement highlights one connection. The map includes historical foundations, enabling methods, and RSI-oriented systems. Logos denote selected author or work-done affiliations, rather than base models; other collaborating institutions may be omitted. The accompanying source register records each selection.

Research routes toward recursive self-improvement

Selected milestones by first public year | Historical roots and the foundation-model era

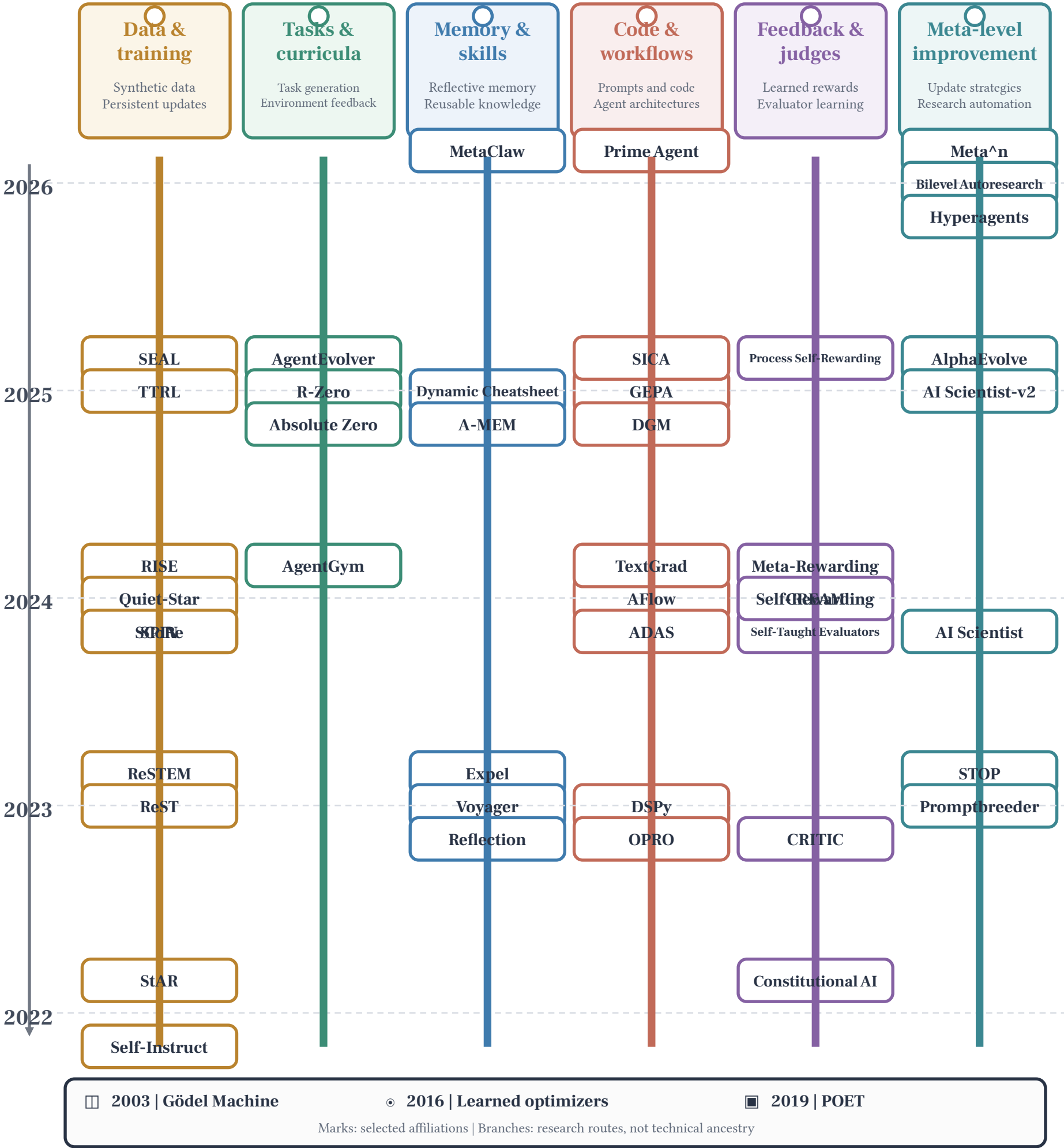


Figure 4. Research routes toward recursive self-improvement. Forty-five representative method papers are grouped into six research routes, with three historical anchors shown separately.

Year bands use the first public dates in the checked source records; spacing within each band is adjusted for legibility. Branch lines denote research families.

Technical ancestry, cross-route transfer, and sustained recursive gains require separate evidence. The selection reflects the literature search through September 12, 2026.

Logos identify one selected author or work-done affiliation per study; “Ind.” denotes independent researchers. Affiliation and logo sources are documented separately.

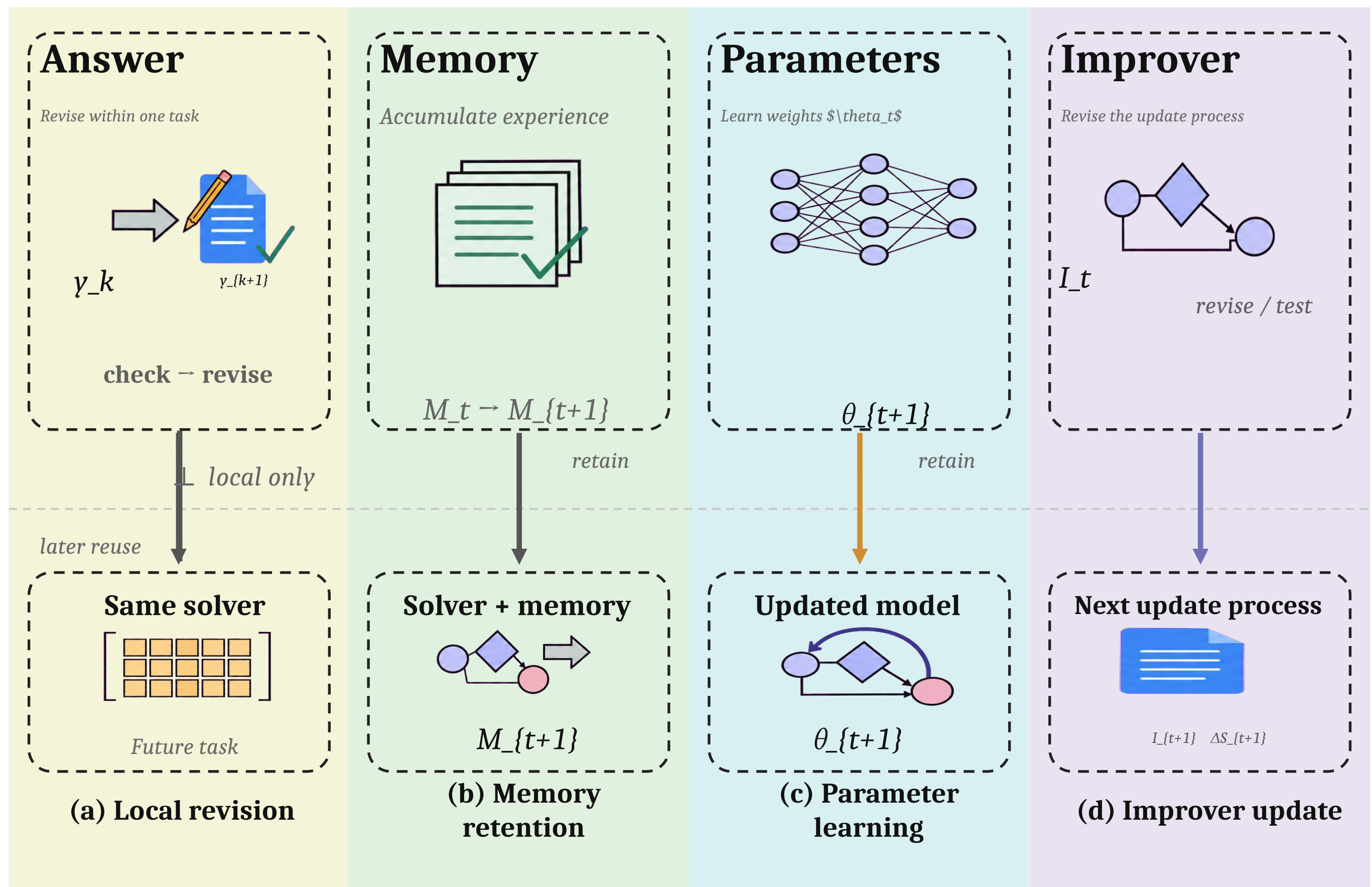


Figure 6. What survives a round of improvement? Four patterns distinguished by the state carried forward. (a) Local checking

and revision change an answer without retaining a change to the solver. (b) Memory M is retained while the model parameters and the chosen update procedure are held fixed. (c) Parameters θ are updated under a fixed training procedure. (d) An accepted change to the improver I is retained and governs a subsequent task-system update ΔS_{t+1} . The horizontal gray dashed line separates current-task or current-round processing from later reuse. Black dashed enclosures group components. Colored arrows denote retained updates, and a capped line denotes a change that is not carried forward. Networks, memory stacks, and procedure graphs are schematic representations. Solver and improver are functional roles that may share an implementation. These analytical patterns may coexist and do not form a capability ranking; persistent or meta-level changes alone do not establish sustained recursive gains.